

Hocine GUIA | AI / Machine Learning Engineer

Paris – Open to opportunities throughout France | guiahocine@gmail.com | [+33 651 98 99 59](tel:+33651989959) | in@guia-hocine | git@Guia-Hocine

PROFILE

Engineer First ,thrived by problem solving and challenges, addicted to learning. if you are looking for someone that understands modern machine learning from first principles I may be a good fit for you.

PROFESSIONAL EXPERIENCE

Agentic AI & ML Engineer, Free Mar 2026 – Sep 2026

- Fine-tuned Gemma4-E2B to generate ClickHouse SQL queries from natural language and evaluated it on internal analytics use cases (L4-H100).
- Fine-tuned a multi-head BERT classifier for internal workflows, then deployed the model through an API for integration into production tools.
- Benchmarked guardrailing techniques and developed a reusable security layer distributable as a package; monitored and improved deployed LLM applications through evaluation and A/B testing.

Key skills: Deep Learning, MCP, Docker, PyTorch, Pydantic, Distillation, Fine-tuning, Bash Scripting, CI

AI Research Intern, Institut Polytechnique de Paris | Télécom SudParis Mar 2025 – Aug 2025

- Collected geospatial data to build knowledge graphs.
- Implemented graph embedding models (TransE, GIE) to encode entities and relations.
- Enhanced spatio-temporal recommender systems (LSTM, TGAT, TGN, JODIE) by integrating spatial knowledge graphs to recommend places to users based on their activity.

Key skills: Graph ML, Knowledge Graphs, Deep Learning, Embedding Models, Web Scraping

AI Engineering & Digital Transformation Consultant, Thynk Tech. | Digital Services Office Sep 2024 – Jan 2025

- Developed an HR platform with LLM agents for résumé analysis and interview simulation, automating HR processes and improving talent management efficiency by 40%.
- Designed a two-portal architecture for candidates and application management.
- Implemented a RAG architecture to leverage internal document repositories.

Key skills: LLMs, RAG, Vector Databases, LangChain, Web Development, Camunda BPMN

Data Science Intern, Amaya AG Feb 2024 – Jun 2024

- Designed a Random Forest-based recommender system for agricultural producers using geospatial and environmental data.

Key skills: Machine Learning, Modelling, Computer Vision, Python, Scikit-learn

EDUCATION

Sorbonne University (formerly UPMC), Master's Degree in Artificial Intelligence (MIND) Sep 2025 – Sep 2026
Master's programme in state-of-the-art AI (Generative Models, PINNs, Autoregressive Models, GANs, GNNs, AI Interpretability)

École Nationale Polytechnique, Dual Degree, Engineering Degree in Data Science & AI Sep 2022 – Jul 2025
Engineering curriculum with a specialisation in data science and artificial intelligence at the country's top-ranked engineering school, offering an advanced learning environment.
Final-year research project in artificial intelligence at Institut Polytechnique de Paris, SAMOVAR Laboratory.

Preparatory Classes for Grandes Écoles, CPGE in Engineering Sciences Sep 2020 – Sep 2022
Preparatory programme in science and mathematics; ranked in the top 1% in the national competitive entrance examination for engineering schools.

PROJECTS, AWARDS AND PUBLICATIONS

Research Paper, SAMOVAR Laboratory, Institut Polytechnique de Paris

- Contributed to the study “*A Context-Aware Spatio-Temporal Model for Human Mobility Forecasting*” at **Institut Polytechnique de Paris**, accepted as a long paper at the IEEE/WIC International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT 2025).

TECHNICAL SKILLS

- **Programming:** Python (NumPy, Pandas, Seaborn, PyTorch, TensorFlow), C++, SQL
- **MLOps:** Git, CI
- **Development:** FastAPI, Neo4j, Streamlit, Docker, Linux, Shell Scripting
- **Agentic AI:** LangChain, MCP
- **Computer Systems:** Networks, Memory, Parallelism